秀傳醫療財團法人彰濱秀傳紀念醫院
Chang Bing Show Chwan Memorial Hospital

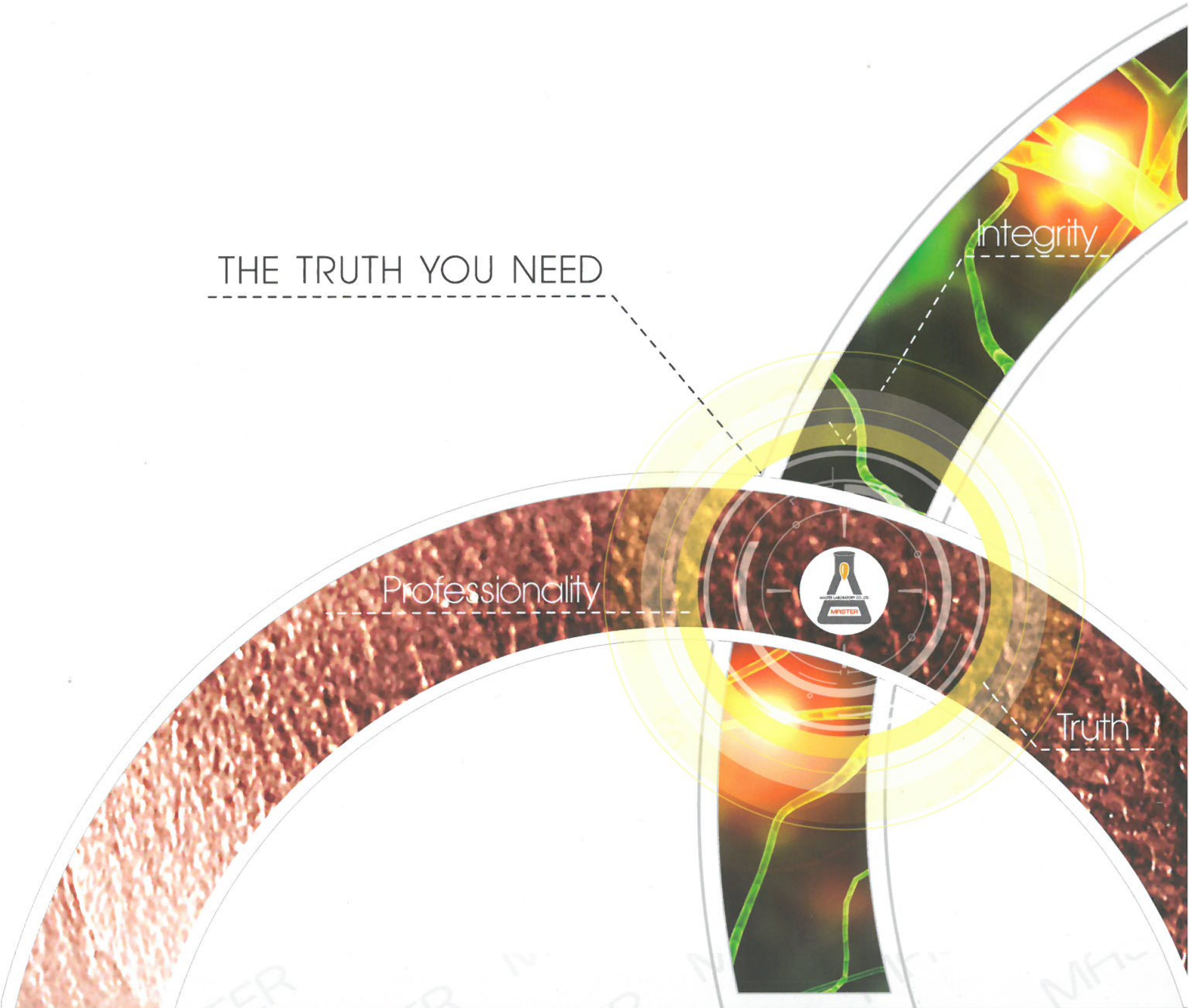
檢測報告

Test Report

Chang Bing Show Chwan

瑞德生物科技有限公司 MASTER LABORATORY CO.,LTD.

THE TRUTH YOU NEED



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瑞德生物科技有限公司
MASTER LABORATORY CO.,LTD.

Pyrogen Study in Rabbits

Master Laboratory Co., Ltd. Animal Laboratory

Address: 3F, No. 221, Sec. 1, Zhongxing Rd., Zhudong Township,

Hsinchu County 31053, Taiwan (R.O.C.) TEL: (886-2)25176117

RELANO WOUND SPRAY
Pyrogen Study in Rabbits
STUDY REPORT

Sponsor：Nanoplus Life Biomedical Technology Co., Ltd.

Testing Institution：Master Laboratory Co., Ltd.

July 2023

Report No.: 23S186T09-01-R01

Experimental starting date: 06.20.2023

Test article registration date: 06.16.2023

Animal in-housing: 06.20.2023

Extraction of test article: 06.26.2023

Experimental administration date: 06.29.2023

Body temperature measurement after administration: 06.29.2023

Animal sacrifice date: 06.29.2023

Experimental completion date: 06.29.2023

Study Announcement

1. The study report is valid for the test article used only, and shall not be partly recopied or extracted for another object.
2. The study report is invalid without the endorsement of Master Laboratory Co., Ltd.

SIGNATURE OF STUDY PERSONNEL

Study Director

Bo Han Huang 07.17.2023

Bo Han Huang **Date**

Facility Management

Alan Hsieh 07.17.2023

Alan Hsieh **Date**

GLP COMPLIANCE STATEMENT

The study met with the technical requirements of the protocol, and the applicable guidance, which included the Good Laboratory Practice for Non-Clinical Laboratory Studies (FDA, 21 CFR, Part 58, 2019), OECD Principles of Good Laboratory Practice (No. 1, 1997) and Good Laboratory Practice for Non-Clinical Laboratory Studies (Food and Drug Administration, R.O.C., 2019), except the parts of test article identification and related analyses (21 CFR §58.105, §58.113, FDA; 6.2.2-6.2.5, OECD GLP No.1, 6.2.2-6.2.5, TFDA GLP). To the best of our knowledge, there were no deviations from the approved study plan and no adverse problem that would affect the integrity of this study or the interpretation of the study result.

Study Director

Bo Han Huang

Bo Han Huang

07. 17. 2023

Date

QUALITY ASSURANCE STATEMENT

The Quality Assurance personnel had inspected the conduct of different phases of the ;tudy according to a predetermined study schedule. To the best of our knowledge, there vere no deviations from the study plan and standard operating procedures that would ıffect the integrity of this study. This report has been audited by the QA personnel in ıccordance with the appropriate standard operating procedures of Master Laboratory Co., Ltd. described the methods and procedures used in the study, and the reported results ıccurately reflect the raw data generated during this study.

Listed below are the phases in this study that were audited by the QA personnel and the lates the audits were performed and findings reported to facility management (FM).

nspection record:

Date of inspection	Phase	Date Reported to SD	Date Reported to FM
06.19.2023	Study protocol, Personnel quality and test article. (Study base)	06.19.2023	06.19.2023
07.03.2023	Study system, study report, raw data, and final report. (Study base)	07.03.2023	07.03.2023

Quality Assurance unit in charge

Ying Chun Chen

'ing Chun Chen

07. 17. 2023

Date

Contract Research Organization and Sponsor Information

. CRO:

a. Title: Master Laboratory Co., Ltd.-Animal laboratory

b. Address: 3F, No. 221, Sec. 1, Zhongxing Rd., Zhudong Township, Hsinchu County 31053, Taiwan, R.O.C.

. Study Director:

a. Name: Bo Han Huang

b. Address: 3F, No. 221, Sec. 1, Zhongxing Rd., Zhudong Township, Hsinchu County 31053, Taiwan, R.O.C.

. Study personnel:

a. Name: Kuo Shu Huang

b. Address: 3F, No. 221, Sec. 1, Zhongxing Rd., Zhudong Township, Hsinchu County 31053, Taiwan, R.O.C.

. Quality Assurance personnel:

a. Name: Ying Chun Chen

b. Address: 3F, No. 221, Sec. 1, Zhongxing Rd., Zhudong Township, Hsinchu County 31053, Taiwan, R.O.C.

. Sponsor:

a. Title: Nanoplus Life Biomedical Technology Co., Ltd.

b. Address: 9 F., No. 179, Sec. 2, Tiding Blvd., Neihu Dist., Taipei City 114735, Taiwan (R.O.C.)

CONTENTS

SUMMARY

The present study was performed in compliance with USP 45/NF40:2022 <151> and ISO/TR 21582:2021 to evaluate the pyrogen limits of test article "RELANO WOUND SPRAY" extracts in the New Zealand White Rabbits by single dose (10 ml/kg) injection in ear vein. The body temperatures of rabbits were measured five times after the single dose intravenous injection. The body temperatures of the administered rabbits were increased within acceptable ranges, pyrogen study responses were negative, the difference between the control temperature and the highest temperature among the five-time measurements was not over 0.5℃. According to the results, the test article extracts met the requirements for the absence of pyrogens.

INTRODUCTION

The testing was performed in compliance with USP 45/NF40:2022 <151> and ISO/TR 21582:2021 to evaluate pyrogen limits of the test article extracts in the New Zealand White Rabbits. The body temperatures of rabbits were measured five times after a single dose (10 ml/kg) intravenous injection in ear vein.

MATERIALS AND METHODS

1. Animals

1.1. Species/Strain: New Zealand White Rabbit

1.2. Resource: Hui Jun

1.3. Body weights (Gender): ＞1.5kg (Male)

1.4. Quarantine/acclimation:

Animals were subjected to be quarantined and acclimated before treatment. Veterinarian ensured the animal health status before the treatment.

1.5. Reasons chosen for animal study:

New Zealand white rabbit was proven to be suitable for pyrogen studies, and widely used in single dose pyrogen studies.

1.6. Groups

Group	Dosage of administration (ml/kg)	Gender	Number
Preliminary test	10	Male	3

2. Feeding and care

2.1. Housing: Rabbit room

2.2. Environment:

a. Temperature: 19 ± 3℃

b. Humidity: 50 ± 20%

c. Light Cycle: 12 hours light and 12 hours dark

2.3. Cage and animal number.

a. Quarantine/acclimation: 1 rabbit/cage

b. Study period: 1 rabbit/cage

2.4. Feed

a. Name: Prolab Rabbit Diet

b. Brand: Lab Diet, U.S.A.

c. Way to supply: *ad libitum*

d. Source: PMI Nutrition International, U.S.A.

2.5. Drinking water

a. Sort: RO Water

b. Way to supply: *ad libitum*

3. Individual and group identification

3.1. Individual identification: Tested animals were identified by ear-marking.

3.2. Group identification: Cages were properly labeled for identification including the Study Title/No., Administration, Observation Period, Room No., Cage No., Quantity/cage, Species, Strain, Gender, In House Date, In House Age, Animal ID No., Keeper and Deputy.

4. Test article and extraction vehicle

4.1. Test article: RELANO WOUND SPRAY, Lot. 51223050003

4.2. Extraction vehicle (Saline): CHI SHENG CHEMICAL CORPORATION, Lot. QB176; Solutio Natrii Chloridi Isotonica 0.9% 500 ml

5. Temperature measuring machine:

5.1. Vendor: LUTRON ELECTRONIC ENTERPRISE CO., LTD.

5.2. Brand: PRECISION 0.01 DEGREE THERMOMETER

6. Administration of test article

6.1. Preparation: According to ISO 10993-12:2021 guideline.

Measured the weight of test article, and immersed it in saline for 72±2hr at 37±1°C with constant agitation (100 rpm) with an extraction ratio of 0.2 g/ml. After extraction, the extract was used immediately; the appearance of test article extract was clear, particulate-free and no color changed.

6.2. The test article extract was injected intravenously into the ear vein of each rabbit with a single dosage of 10 ml/kg.

6.3. The injection volume of test article extract was based on body weights which recorded on the first day of study prior to administration.

7. Procedure

7.1. Three days before the start of the experiment, body temperatures of the 3 rabbits were measured for sham test. The criteria of sham test which rabbits selected for pyrogen study is the same as paragraph 7.4 and 8.1.

7.2. Inserted the temperature measuring machine probe into the rectum of the test rabbit to a depth of not less than 7.5 cm and measured with a constant time of 1 minute.

7.3. During the study, animals were fasting and only RO water was provided. Thirty minutes prior to the injection of the test article extract, the body temperatures of each animal were measured using anus thermometer (accuracy ± 0.01℃). The temperature measured 30 minutes before the injection is defined as the control temperature. Every appliance used in the study was also pyrogen-free.

7.4. Three rabbits whose control temperatures did not vary by more than 1°C from each other were collected for use. Any rabbit with a control temperature exceeding 39.8°C was excluded. Test article extract (warm up to 37 ± 2℃) was injected into ear vein of the rabbit. The time of administration was not over than 10 minutes. Body temperatures were measured subsequently to the administration at 30-minute intervals between 1 and 3 hours. The differences of the highest body temperature among the five-time measurements and the control temperature were evaluated.

8. Criteria

8.1. If the difference between the control temperature and the highest temperature among the five-time measurements was not over 0.5℃, the test article would meet the requirements for the absence of pyrogen.

9. Archive of the record and data

9.1. All records and data were preserved for six years. Test articles (retained samples) were retained for six years excluding the samples that sponsors require to be fully returned. Archive address: 3F., No. 221, Sec. 1, Zhongxing Rd., Zhudong Township, Hsinchu County 31053, Taiwan (R.O.C.).

RESULTS

1. Body weight (Table 2-2)

Body weights of three animals were above 1.5 kg and were qualified for the study.

2. Body temperature of rabbits before study day (Table 1-1, 1-2, 1-3)

The sham test measurements of each animal before the study day did not exceed 39.8 degrees and the difference between any one group of test rabbis did not exceed 1℃ and the elevation of body temperatures of the three rabbits were below 0.5℃, indicating that the selected animals met the requirements states in paragraph 7.1.

3. Control temperature (Table 3-1)

Control temperatures in three animals were 39.44℃, 39.51℃ and 39.42℃, respectively. Those all not exceeded 39.80℃ and the temperature difference between those animals did not exceed 1℃, which met the criteria.

4. Body temperatures measurement after test article administration (Table 3-2).

5. The elevation of body temperatures of the three rabbits were below 0.5℃ (Table 2-1). (Animal No. 1001: 0.12℃, Animal No. 1002: 0.04℃ and Animal No. 1003: 0.09℃)

CONCLUSION

The study was performed in compliance with USP 45/NF40:2022 <151> and ISO/TR 21582:2021 to evaluate the pyrogen limits of test article extracts in the New Zealand White Rabbits by single dose (10 ml/kg) injection in ear vein. The body temperatures of rabbits were measured five times after the single dose intravenous injection. The response of pyrogen study was negative; therefore, the test article "RELANO WOUND SPRAY" extracts under examination met the requirements for the absence of pyrogens.

REFERENCES

1. Good Laboratory Practice for Nonclinical Laboratory Studies (2019) Food and Drug Administration, R.O.C.

2. Good Laboratory Practice for Nonclinical Laboratory Studies. Title 21 of the U.S. Code of Federal Regulations, Part 58 (2019) United States Food and Drug Administration.

3. Pharmacopeia US: Pyrogen Test, USP 45/NF40:2022 <151>.

4. Pyrogenicity-Principles and methods for pyrogen testing of medical devices, ISO/TR 21582:2021.

5. Biological evaluation of medical devices-Part 1: Evaluation and testing within a risk management process ISO 10993 (2018)

6. Biological evaluation of medical devices-Part 11: Tests for systemic toxicity ISO 10993 (2017).

7. Biological evaluation of medical devices-Part 12: Sample preparation and reference materials ISO 10993 (2021).

8. Biological evaluation of medical devices-Part 12: Sample preparation and reference materials EN ISO 10993 (2021).

9. General requirements for the competence of testing and calibration laboratories. ISO/IEC 17025 (2017).

Table 1-1. Control Temperature of Sham Test

Animal number	Control temperature (℃)
1001	39.38
1002	39.44
1003	39.36

Remark: Measured body temperature readings recorded for the rabbit in 30 minutes before the administration. The body temperature was the control temperature.

Table 1-2. Body Temperature Elevation of Sham Test

Animal number	Body temperature elevation (℃)
1001	0.04
1002	0.08
1003	0.03

Remark: Temperature elevation: the highest body temperature (Among five-time measurement) subtract control temperature.

Table 1-3. Body Temperature Records (Sham Test)

Animal number	Body temperature (℃)					
	1st	2nd	3rd	4th	5th	HBT-CT
1001	39.25	39.31	39.40	39.33	39.42	0.04
1002	39.40	39.37	39.45	39.46	39.52	0.08
1003	39.31	39.34	39.39	39.35	39.28	0.03

Remark: Body temperatures were measured 1, 1.5, 2, 2.5 and 3 hr after administration for five times at half-hour intervals.

HBT: highest body temperature

CT: control temperature

Table 2-1. Body Temperature Elevation

Animal number	Body temperature elevation after administration (℃)
1001	0.12
1002	0.04
1003	0.09

Remark: Temperature elevation: the highest body temperature (Among five-time measurement) subtract control temperature.

Table 2-2. Individual Body Weight

Dosage (ml/kg)	Gender	Animal number	Weight (g)	Injection Volume (ml)
10	Male	1001	2227	22
		1002	2165	22
		1003	2190	22

Table 3-1. Control Temperature of rabbits

Animal number	Control temperature (℃)
1001	39.44
1002	39.51
1003	39.42

Remark: Measured body temperature readings recorded for the rabbit in 30 minutes before the administration. The body temperature was the control temperature.

Table 3-2. Body Temperature Records (After Administration)

Animal number	Body temperature (℃) after administration					
	1st	2nd	3rd	4th	5th	HBT-CT
1001	39.39	39.43	39.48	39.56	39.50	0.12
1002	39.47	39.42	39.53	39.55	39.48	0.04
1003	39.35	39.38	39.46	39.43	39.51	0.09

Remark: Body temperatures were measured 1, 1.5, 2, 2.5 and 3 hr after administration for five times at half-hour intervals.

HBT: highest body temperature

CT: control temperature

Appendix 1. Test Article Information Sheet

Master Laboratory Co., Ltd.

Information for ■Test Article/□Control Article

Sponsor Company	Nanoplus Life Biomedical Technology Co., Ltd.
Sponsor Address	9 F., No. 179, Sec. 2, Tiding Blvd., Neihu Dist., Taipei City 114735, Taiwan (R.O.C.)
Contract Study item	■Base on the contract
ame of test article	RELANO WOUND SPRAY (良諾液態保護噴劑)
Major omponents	Nano Ion Water, Sodium Carboxymethylcellulose
Sample status	□Sterilized (□Gamma □EO□Steam) ■Not Sterilized
Storage condition	■Room temperature (10°C-30°C) □Refrigeration (2°C-8°C) □Others:
Expiry day	05 / 26 / 2025
Specific equirement	None.
Lot number	■Base on the specific number on the package: LOT51223050003 □Others:
Extract by	■ Weight (0.2g/ml) Total weight of each test article: about 50 g (1g/ml) □ Surface (Sample thickness: □>1.0mm □0.5-1.0mm□<0.5mm) Total area surface of each test article:
Absorption	■ Non absorption □ Water absorption rate:/ Oil absorption rate:
Sponsor Signature	[illegible] 06/15/2023

Appendix 2. Test Article

MASTER LABORATORY CO.,LTD.

MASTER

秀傳紀念醫院
Show Chwan Memorial Hospital
彰化市中山路一段542號 04-7256166
No.542, Sec. 1, Zhongshan Rd., Changhua City,
Changhua County 500, Taiwan (R.O.C.)

彰濱秀傳紀念醫院
Chang Bing Show Chwan Memorial Hospital
彰化縣鹿港鎮鹿工路6號 04-7813888
No.6, Lugong Rd., Lukang Township,
Changhua County 505, Taiwan (R.O.C.)

秀傳亞洲遠距微創手術中心
Asian Institute of TeleSurgery
彰化縣鹿港鎮鹿工路6-1號 04-7812988
No.6-1, Lugong Rd., Lukang Township,
Changhua County 505, Taiwan (R.O.C.)

洪宗鄰醫院
Hung Tsung Lin Hospital
彰化縣二林鎮中正路61號 04-8967955
No. 61, Zhongzheng Rd., Erlin Township,
Changhua County 526, Taiwan (R.O.C.)

田中仁和醫院
Tien Chung Jen Ho Hospital
彰化縣田中鎮中州路一段157號 04-8742108
No.157, Sec. 1, Zhongzhou Rd., Tianzhong Township,
Changhua County 520, Taiwan (R.O.C.)

台北秀傳醫院
Taipei Show Chwan Hospital
台北市光復南路116巷1號 02-27717172轉66
No.1, Ln. 116, Guangfu S. Rd., Songshan Dist.,
Taipei City 105, Taiwan (R.O.C.)

竹山秀傳醫院
Chu Shang Show Chwan Hospital
南投縣竹山鎮集山路二段75號 049-2624266
No.75, Sec. 2, Jishan Rd., Zhushan Township,
Nantou County 557, Taiwan (R.O.C.)

台南市立醫院
Tainan Municipal Hospital
台南市東區崇德路670號 06-2609926
No.670, Chongde Rd., East Dist.,
Tainan City 701, Taiwan (R.O.C.)

高雄縣立岡山醫院
kaohsiung Municipal Gangshan Hospital
run by Show Chwan Memorial Hospital
高雄縣岡山區壽天路12號 [illegible]
No.12, Shoutian Rd., Gangshan Dist.,
Kaohsiung City 820, Taiwan (R.O.C.)

秀傳醫療財團法人彰濱秀傳紀念醫院
Chang Bing Show Chwan Memorial Hospital

地址：505彰化縣鹿港鎮鹿工路6號
電話：04-7813888
網址：www.cbshow.org.tw/